Arun Bagavathiannan

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Education

The University of Texas at Austin — Austin, TX
Bachelor of Science, Computer Science — *Expected Spring 2028*

- Relevant Coursework: CS 311 (Discrete Math), CS 314 (Data Structures), ECE 312H (Software Design & Implementation I), ECE 319H (Embedded Systems), M 340L (Matrices and Matrix Operations)

Experience

Technical Lead, UT Registration Plus — Jan. 2026 – Present
Longhorn Developers

- Promoted from Software Engineer to Technical Lead in Aug. 2026; leading a team of 6 developers on a Chrome extension serving 60,000+ students
- Ran end-to-end hiring for the developer fellowship: screened 80+ applications, interviewed finalists, and selected 5 engineers (~6% acceptance)
- Engineered a Course Status Checker that enables real-time course availability tracking within the calendar UI — cited in UT CS News coverage of the extension
- Designed a throttled refresh system in React and TypeScript to prevent API rate-limiting while maintaining compliance with university infrastructure constraints
- Built a reactive store architecture to synchronize state between background service workers and the frontend UI

Content Creator — Jul. 2021 – Present
YouTube

- Grew YouTube channel to over 20,000 subscribers with 32M+ impressions and 7M+ views by shipping consistent, high-quality video editing content through After Effects and Alight Motion
- Launched and operated buyorphic.com, an e-commerce site selling editing presets, generating $500+ in sales over one year

ML Research Intern — Jan. 2025 – Mar. 2025
Texas A&M University

- Presented results at 3 conferences including the Weed Science Society of America (Vancouver) and European Weed Research Society (Spain)
- Designed and trained a YOLOv8 object detection pipeline, taking a raw dataset of 100+ images to a production-ready model with 80%+ precision and recall
- Engineered automated data augmentation workflows using Python and OpenCV, scaling training data 6x (100 to 600+ images) and cutting manual processing time

Projects

Investly — Feb. 2026 – May 2026

- Built end-to-end LLM research pipeline in Python: RSS ingestion, local Gemma 3 4B classification, ChromaDB RAG, autonomous daily paper trading against SPY
- Measured news coverage vs forward returns (=+0.34, p=0.001 at 3d, inverting by 1m); backtested and rejected the resulting strategy

Skills

Programming: Python, Bash, Assembly, C, C++, TypeScript, Java
Tools/Platforms: ChromaDB, Chrome Extension (Manifest V3), React, Node.js, Git/GitHub
AI/ML: Gemma 4 E4B, LLM Inference, RAG Pipelines, Model Training/Validation, MCP, Claude API, Ollama
Languages: English (Native), Tamil (Native), German (Novice), Japanese (Novice)

Awards

1st Place, AI Data Portability Hackathon – The AI Collective, AITX, DTI — Mar. 2026
International Education Fee Scholarship (IEFS) Recipient — Mar. 2026
Friends of Alec Engineering Scholarship Recipient — Feb. 2026